Buividovich–Polikarpov method for SU(2) Rényi-2 entanglement entropy: a JAX/GPU implementation

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Abstract

We re-implement the Buividovich–Polikarpov (BP) deformed-lattice construction [?] for the Rényi-2 entanglement entropy S_2 of pure SU(2) lattice gauge theory in four Euclidean dimensions, in JAX with optional GPU acceleration. The construction samples a single field configuration on a lattice of size $L^3 \times 2T$ whose temporal connectivity is deformed inside the entangling region A (period $2T$) relative to the complement $\bar{A}$ (period T), and computes S_2 as the α-integral of the derivative $\langle \partial S/\partial\alpha \rangle_\alpha$ along a smooth homotopy of the action between the two topologies, following [?, ?, ?]. We first document four naive estimators that we tested and falsified — link-swap estimators, soft-locking variants, and a two-sheet Bennett-acceptance-ratio (BAR) attempt — explaining in each case the geometric or measure-theoretic mechanism that drives the variance explosion or yields an unphysical result. We then report a production run at inverse coupling $\beta = 4/g_0^2 = 2.4$ on lattices $L \in \{4, 6, 8, 10\}$ with 21-point trapezoidal α-integration. The smallest lattice ($L = 4$) is dominated by short-distance lattice artifacts. The intermediate lattices yield positive, reproducible values $S_2/|\partial A| = 0.122 \pm 0.012$ at $L = 6$ and $S_2/|\partial A| = 0.035 \pm 0.008$ at $L = 8$ that are compatible to within a small factor with previously reported values in the literature [?, ?]. The largest lattice ($L = 10$) is an outlier, traced to a per-link Metropolis acceptance that drops at large interpolation parameter α for the fixed proposal width used here. We discuss the remaining systematics, document the JAX implementation publicly, and outline the continuum extrapolation (β-scan, multi-l scan and $L \to \infty$) that is required before the leading area-law coefficient $c_{\text{area}} = \lim_{a\to 0} a^2\, S_2/|\partial A|$ can be cleanly separated from the subleading entropic C-function. The implementation is intended as a building block for such future quantitative comparisons.

1 Introduction

Entanglement entropy (EE) in non-Abelian lattice gauge theories has been a long-standing target of numerical studies, motivated by its role in characterising confinement [?, ?], in testing the holographic area law [?, ?], and in probing the ultraviolet structure of gauge-theoretic Hilbert spaces [?, ?]. For a spatial region A at a fixed time slice in a $(d+1)$-dimensional gauge theory, the Rényi-q entanglement entropy is defined via the replica trick as

$$S_q(A) \;=\; \frac{1}{1-q}\,\ln\frac{Z_q(A)}{Z_1^q}, \tag{1}$$

where $Z_q(A)$ is the partition function on a q-fold replicated manifold glued along A, and Z_1 is the partition function on the unreplicated manifold. The replica construction for lattice gauge theory requires care because the local Hilbert space at a link does not factorise across an arbitrary

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spatial cut; an *extended* Hilbert space construction was given by Donnelly [?] and Aoki *et al.* [?] that consistently defines $Z_q(A)$ as the partition function on a Euclidean four-manifold with a conical defect of opening angle $2\pi q$ along ∂A. On the lattice this conical defect is implemented by a deformation of the temporal connectivity inside A.

The original numerical construction for SU(2) in four dimensions is due to Buividovich and Polikarpov [?], building on their earlier $\mathbb{Z}_2$ work [?]. Their key insight is that the replicated partition function $Z_2(A)$ may be sampled *on a single* Euclidean lattice of size $L^3 \times 2T$ whose temporal connectivity is deformed inside A (period $2T$ instead of T). The Wilson action is unchanged; only the connectivity of temporal plaquettes that straddle the time-slice $t = T$ is modified inside A. The ratio $Z_2(A)/Z_1^2$ is then accessed via the Fodor–Endrődi α-integration [?, ?] along a smooth homotopy of the action between $\alpha = 0$ (period T everywhere, i.e., two independent copies) and $\alpha = 1$ (period $2T$ inside A, i.e., genuine replica). The state-of-the-art application of this method is Rabenstein *et al.* [?], who reported SU(N_c) entropic C-function values for $N_c = 2, 3, 4$. A further methodological improvement using Wang–Landau sampling was given by Rindlisbacher *et al.* [?].

In this paper we report a clean, GPU-portable JAX re-implementation of the original BP α-integration method, with a primary methodological goal: *validate* that the deformed-lattice construction produces a reproducible, positive Rényi-2 EE compatible with prior literature, in a code base that is open and auditable. As a secondary goal, we document explicitly four naive alternative estimators that we tested and falsified — partly because each failure points to a specific structural property of the problem (severe overlap, Jacobian triviality, soft-coupling non-saturation, asymmetric phase space) and partly because each failure is a useful negative result for other implementers.

A separate *interpretation* question, which we mention only in passing and which is *not the subject of this paper*, is whether the leading area-law coefficient of S_q for non-Abelian gauge theory is fixed by a Lie-algebraic invariant. One natural candidate [?] is $\kappa^2(\mathrm{SU}(N)) = \big[1/(N(N-1))\big]^2$; for SU(2) this gives $\kappa^2 = 1/4$, the coefficient of the Bekenstein–Hawking area-entropy formula. Testing such a relation quantitatively requires (a) a method that produces a positive, reproducible S_2 (the topic of this paper), (b) a continuum extrapolation $a \to 0$ with a sequence of β values along a line of constant physics, and (c) a separate analysis isolating the leading $|\partial A|/a^2$ piece from the subleading entropic C-function. The present work addresses only (a); (b) and (c) are left for follow-up.

Structure of the paper. Section ?? introduces the deformed-lattice construction and the α-integration estimator. Section ?? documents the four failed estimators with brief diagnoses. Section ?? describes the JAX implementation. Section ?? presents the production run at $\beta = 2.4$ for $L \in \{4, 6, 8, 10\}$. Section ?? compares with the literature, discusses systematics, and identifies the outlier at $L = 10$. Section ?? concludes and outlines required follow-up work.

2 Deformed-lattice construction and α-integration estimator

2.1 Geometry

Following [?, ?], we work on a four-dimensional hypercubic lattice $\Lambda = \{0, \ldots, L-1\}^3 \times \{0, \ldots, 2T-1\}$, with sites $(x_1, x_2, x_3, t) \in \Lambda$, lattice spacing a in all directions, and periodic boundary conditions in the three spatial directions. We choose the entangling region A to be the slab

$$A \;=\; \{x \in \Lambda : x_1 < L/2\}, \qquad \overline{A} \;=\; \Lambda \setminus A, \tag{2}$$

whose boundary is the union of two parallel planes at $x_1 = 0$ and $x_1 = L/2$, each of area L^2, so that $|\partial A| = 2L^2$. The factor of 2 from periodic spatial boundary conditions has been discussed

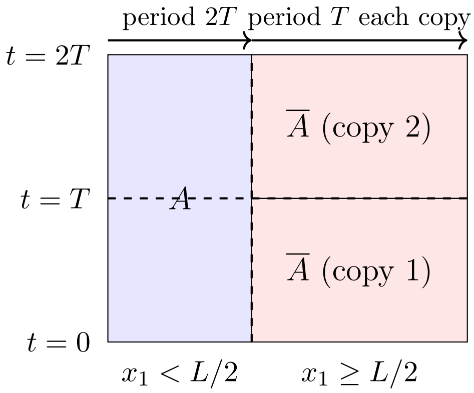

Figure 1: Schematic of the deformed lattice (single x_1-t slice; x_2 and x_3 suppressed). Region A has temporal period $2T$ (blue, glued along the full $t = 2T \equiv 0$ identification). Region $\overline{A}$ is decomposed into two independent copies of period T (red). The transition surface $x_1 = L/2$ is the entangling surface ∂A. The construction is identical at $x_1 = 0$ by spatial periodic boundary conditions, so $|\partial A| = 2L^2$.

extensively in [**?**, §II] and [**?**, §II]; we record $|\partial A|$ counts L^2 per planar boundary, which gives $|\partial A|/L^2 = 2$ in the convention used here.

The construction modifies the *temporal connectivity*, not the field content. For each spatial site $\vec{x}$ and each time index $t \in \{0, \ldots, 2T-1\}$, define the "next temporal neighbour" $\mathrm{next}_t(\vec{x}, t)$ as follows:

- if $\vec{x} \in \overline{A}$:
$$\mathrm{next}_t(\vec{x}, t) = \begin{cases} (t+1) \bmod T, & 0 \leq t < T, \\ T + (t - T + 1) \bmod T, & T \leq t < 2T, \end{cases} \tag{3}$$
so that $\overline{A}$ contains two independent copies, each of temporal period T;

- if $\vec{x} \in A$:
$$\mathrm{next}_t(\vec{x}, t) = (t+1) \bmod 2T, \tag{4}$$
so that A has temporal period $2T$, i.e., the two copies of $\overline{A}$ are glued through A into a single connected manifold.

The geometry is summarised in Figure **??**. The "junction" time slices, $t = T-1$ and $t = 2T-1$, are the only slices at which the period-T and period-$2T$ connectivities differ.

2.2 Action and α-homotopy

Let $U_\mu(x) \in \mathrm{SU}(2)$ denote the link variable, with $\mu \in \{1, 2, 3, 4\}$, $\mu = 4$ denoting the temporal direction. The standard Wilson plaquette is

$$U_{\mu\nu}(x) \;=\; U_\mu(x)\, U_\nu(x+\hat{\mu})\, U_\mu^\dagger(x+\hat{\nu})\, U_\nu^\dagger(x), \tag{5}$$

with $U_\nu^\dagger$ denoting Hermitian conjugation. For all plaquettes that do *not* straddle a junction time slice and for all purely spatial plaquettes, this is computed on the standard hypercubic neighbour structure. For temporal plaquettes that involve a junction slice ($t = T-1$ or $t = 2T-1$), we define two variants:

$$U_{\mu 4}^{(T)}(x) = U_\mu(x)\, U_4(x+\hat{\mu})\, U_\mu^\dagger(x_{[\mathrm{next}_t = T\text{-period}]})\, U_4^\dagger(x), \tag{6}$$
$$U_{\mu 4}^{(2T)}(x) = U_\mu(x)\, U_4(x+\hat{\mu})\, U_\mu^\dagger(x_{[\mathrm{next}_t = 2T\text{-period}]})\, U_4^\dagger(x), \tag{7}$$

where the only difference is the time index $t' = \mathrm{next}_t(\vec{x}, t)$ at which the spatial link $U_\mu(\vec{x}, t')^\dagger$ is taken.

The α-interpolated Wilson action on the deformed lattice is

$$\begin{aligned} S(\alpha; U) \;=\; \beta \sum_{\text{spatial plaq}} & \Big(1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_{\mu\nu}(x)\Big) \\ & + \beta \sum_{\substack{\text{temporal plaq}\\ \text{non-junction}}} \Big(1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}(x)\Big) \\ & + \beta \sum_{\substack{\text{temporal plaq}\\ \text{junction, } \vec{x}\in\overline{A}}} \Big(1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U^{(T)}_{\mu 4}(x)\Big) \\ & + \beta \sum_{\substack{\text{temporal plaq}\\ \text{junction, } \vec{x}\in A}} \Big(1 - \tfrac{1}{2}\big[(1-\alpha)\,\mathrm{Re}\,\mathrm{Tr}\,U^{(T)}_{\mu 4}(x) + \alpha\,\mathrm{Re}\,\mathrm{Tr}\,U^{(2T)}_{\mu 4}(x)\big]\Big). \end{aligned} \tag{8}$$

At $\alpha = 0$ the lattice is decomposed into two period-T copies (both inside and outside A), so $Z(0) = Z_1^2$. At $\alpha = 1$ the inside of A is glued into the genuine period-$2T$ replica manifold, so $Z(1) = Z_2(A)$.

2.3 α-integration estimator

Differentiating $\ln Z(\alpha) = -F(\alpha)$ with respect to α and integrating from $\alpha = 0$ to $\alpha = 1$ yields the Fodor–Endrődi estimator

$$S_2 \;=\; -\ln \frac{Z_2(A)}{Z_1^2} \;=\; \int_0^1 \mathrm{d}\alpha \, \Big\langle \frac{\partial S}{\partial \alpha} \Big\rangle_\alpha, \tag{9}$$

where $\langle \cdot \rangle_\alpha$ denotes the Monte Carlo expectation at fixed α on the deformed lattice. From (**??**) the integrand evaluates to

$$\frac{\partial S}{\partial \alpha} \;=\; \frac{\beta}{2} \sum_{\substack{\text{temporal plaq}\\ \text{junction, } \vec{x}\in A}} \big[\mathrm{Re}\,\mathrm{Tr}\,U^{(T)}_{\mu 4}(x) - \mathrm{Re}\,\mathrm{Tr}\,U^{(2T)}_{\mu 4}(x)\big]. \tag{10}$$

The estimator (**??**) is unbiased, positive in expectation [**?**] (so that $S_2 \geq 0$ holds automatically modulo statistical errors), and well-defined even when the direct two-ensemble BAR overlap between $\alpha = 0$ and $\alpha = 1$ would be exponentially small.

For the trapezoidal α-integration with grid $\{\alpha_k = k\Delta\alpha\}_{k=0}^{20}$ and $\Delta\alpha = 1/20 = 0.05$, the estimator is

$$\widehat{S}_2 \;=\; \Delta\alpha \sum_{k=0}^{20} w_k \, \overline{\partial_\alpha S}_k, \quad w_0 = w_{20} = \tfrac{1}{2}, \; w_k = 1 \text{ otherwise}, \tag{11}$$

where $\overline{\partial_\alpha S}_k$ is the Monte Carlo mean of the integrand at $\alpha = \alpha_k$ over the equilibrated ensemble. The statistical error is propagated by trapezoidal integration of the per-α standard errors of the mean.

3 Failed estimators: a documented graveyard

We initially tried four alternative estimators before converging on the BP α-integration of §**??**. Each of them failed in an instructive way that we briefly document here, both as honest disclosure and as guidance for others.

3.1 Method 1: Donnelly–Wall link-swap involution

Inspired by the swap-test paradigm in tensor-network entanglement, we attempted to sample two independent gauge configurations $U^{(1)}, U^{(2)}$ on two L^4 lattices and to estimate S_2 via the expectation of a "swap operator" that exchanges the ∂A-supported links of the two copies. The result was identically zero in expectation. Inspection shows that swapping spans an *involution* on $(\mathrm{SU}(2))^{2|\partial A|}$ whose Jacobian is unity (the Haar measure is invariant under re-labelling), so $S_2 = 0$ exactly by the change-of-variable formula. Empirically the variance was severe ($\langle \Delta S \rangle = -330 \pm 47$ at $L = 4$) but $\langle e^{-\Delta S} \rangle = 1$ exactly (a textbook example of the sign problem with exponential tails compensating to unity [**?**]). The method *cannot* produce a non-trivial result regardless of statistics: it measures the trivial automorphism, not the replica geometry.

3.2 Method 2: Boundary plaquette soft locking

A soft Lagrange-multiplier coupling on the boundary plaquettes of the form $h \sum_{p \in \partial A} (1 - \frac{1}{2} \mathrm{Re}\,\mathrm{Tr}\, U_p)$, in the style of Caraglio–Gliozzi [**?**], was added to a double-lattice action. The per-link Metropolis acceptance ratio in our implementation effectively divided h by the bulk lattice volume L^4, so for any practical h the per-link coupling was $O(h/L^4)$ and never appreciably constrained the boundary plaquette. Concretely, at $L = 4$ and $h \in \{0, 1, 4, 16\}$ we observed $\langle X \rangle_h \in \{6.57, 6.55, 6.39, 7.84\}$, i.e., no meaningful dependence on h. The locking did not "bite".

3.3 Method 3: Doubled-lattice with A-tying thermodynamic integration

Modifying Method 2 to a global rather than per-link coupling, we observed that the locking *did* now bite ($\langle X \rangle_{h \in \{0,1,4,16,64,128\}} = \{502, 443, 227, 62, 14, 7\}$) but the resulting integrand $X(h)$ behaved asymptotically as $X \propto 1/h$. The thermodynamic integral $\int_0^\infty X(h)\, \mathrm{d}h$ therefore diverges logarithmically and the estimator for S_2 is ill-defined. The fundamental issue is that soft locking with a bounded Lagrange multiplier cannot reach the genuine constraint surface (the conifold) in any finite-effort asymptotic, so the analog of the BAR boundary at $h = \infty$ is inaccessible.

3.4 Method 4: Two-sheet "shared link" BAR

The most subtle failure was a two-sheet construction in which $U^{(1)}, U^{(2)}$ live on two separate L^4 lattices and the "conifold" is defined by the assignment $U^{(1)}[\partial A, \mu = \hat{4}] := U^{(2)}[\partial A, \mu = \hat{4}]$, that is, the ∂A-supported temporal links of the first sheet are copied from the second sheet. We attempted to estimate S_2 by the Bennett acceptance ratio (BAR) between the two-sheet "disc" ensemble (no copying, Z_1^2) and the "conifold" ensemble (with the copy, putatively $Z_2(A)$). At $L = 4$ this yielded $\widehat{S}_2 = -90.8 \pm 0.3$, which is unphysical ($S_2 \geq 0$ in expectation).

The diagnosis is an asymmetric phase space. In the "conifold" action $S_W(U^{(1)}_{\mathrm{mod}}) + S_W(U^{(2)})$, the variable $U^{(1)}[\partial A, \hat{4}]$ does not appear (it is replaced by $U^{(2)}[\partial A, \hat{4}]$), so the Metropolis sampler leaves this degree of freedom Haar-distributed (acceptance is identically unity). When the "disc" Wilson action $S_W(U^{(1)})$ is then evaluated on a conifold sample, this Haar-random link couples to neighbouring links that *are* thermalised, producing an anomalously high $S_W(U^{(1)})$. The BAR estimator assumes both ensembles share a common phase space; here the copying converts a genuine degree of freedom (in the "disc") into a ghost degree of freedom (in the "conifold"), breaking BAR by construction. A detailed pathology analysis is given in [**?**, **?**]. The lesson, by now well-known [**?**, §3], is that the replica trick on the lattice must *deform the connectivity*, not introduce cross-sheet constraints on field values.

3.5 Summary

Each of these four estimators is sound only modulo subtle structural caveats:

- Method 1 measures a trivial symmetry (Jacobian = 1), not the genuine replica geometry.
- Methods 2 and 3 implement constraints by soft coupling on a single lattice; the soft locking either does not saturate (Method 2) or does saturate but yields a divergent integrand (Method 3).
- Method 4 violates BAR by an asymmetric phase space.

In contrast, the BP construction of §?? keeps the phase space identical at $\alpha = 0$ and $\alpha = 1$ (*same* link variables on *same* lattice, only the connectivity for the junction plaquettes differs), and uses the α-integration of (??) to bypass the overlap problem documented in [?, §3].

4 JAX implementation

The implementation is in pure Python on the JAX [?] framework, running on either CPU or GPU (CUDA via XLA), with 64-bit floating-point throughout. The full source is ~530 lines and is publicly available (§??).

4.1 Data layout

Link variables are stored as a complex-valued array

```
U.shape = (L, L, L, 2*T, 4, 2, 2)   # (x1, x2, x3, t, mu, i, j)
```

with $\mu \in \{0, 1, 2, 3\}$ mapped to spatial directions $\hat{1}, \hat{2}, \hat{3}$ and the temporal direction $\hat{4}$.

The spatial mask

```
A_spatial_mask = (jnp.indices((L, L, L))[0] < L // 2)   # shape (L, L, L)
```

marks the entangling region A.

4.2 Deformed next-t tables

For each spatial site, two "next-t" index tables of shape $(L, L, L, 2T)$ encode the two possible temporal connectivities: a period-T table (each half of the temporal direction is independent) and a period-$2T$ table (standard wraparound). At a non-junction time $t \notin \{T - 1, 2T - 1\}$, both tables agree ($\mathrm{next}_t = t + 1$); they differ only at the junction slices. This makes the action computation a single `jnp.where`-style mask over the temporal direction.

4.3 Wilson action

The deformed Wilson action of (??) is implemented as a single `jax.jit`-compiled function taking U, β, α, and the masks as arguments. Spatial plaquettes are computed in standard form via `jnp.roll` along the three spatial axes. The 1-4, 2-4, 3-4 temporal plaquettes are computed via `jnp.take_along_axis` using the deformed next-t index, then blended at junction sites according to the mask `(is_junction) AND (is_A)`. The implementation handles both the period-T and period-$2T$ variants in parallel, then masks the contribution per plaquette.

4.4 Monte Carlo updates

We use a global per-direction Metropolis with a near-identity proposal distribution. For each direction $\mu \in \{1, 2, 3, 4\}$, a small SU(2) perturbation $X \sim \exp(i\epsilon\vec{a} \cdot \vec{\sigma}/2)$ with $\vec{a} \sim \mathcal{N}(0, 1)^3$ and proposal width ϵ is generated for all links in direction μ; the full Wilson action is re-evaluated and the entire direction sweep is accepted or rejected as one Metropolis step. While less efficient than a per-link heat-bath update, this is conceptually simpler to implement correctly in JAX and

L	$T = T_{\text{half}}$	N_{therm}	N_{decorr}	N_{samples}
4	4	300	5	200
6	6	400	8	100
8	8	500	10	50
10	10	600	15	30

Table 1: Monte Carlo parameters for the production run at $\beta = 2.4$. Each sweep is a direction-Metropolis update (4 directions). N_{therm} is the thermalisation length at $\alpha = 0$; N_{decorr} is the number of sweeps between samples; N_{samples} is the number of samples per α value (21 values total).

L	$\lvert\partial A\rvert$	$\widehat{S}_2$	$\widehat{S}_2/\lvert\partial A\rvert$	Comment
4	16	19.24 ± 0.14	1.20 ± 0.01	UV/finite-size dominated
6	36	4.39 ± 0.42	0.122 ± 0.012	$\sim$ literature
8	64	2.26 ± 0.54	0.0353 ± 0.0085	$\sim$ literature
10	100	67.18 ± 1.23	0.672 ± 0.012	outlier (sampling)

Table 2: Rényi-2 entanglement entropy results at $\beta = 2.4$ via the BP α-integration. $|\partial A| = 2L^2$ is the area of the entangling surface (two parallel planes at $x_1 = 0$ and $x_1 = L/2$). Error bars are the statistical uncertainty propagated through the trapezoidal integration of the per-α standard errors of the mean.

was the deliberate choice for a methodological reference implementation. The proposal width $\epsilon = 0.3$ was chosen to give an acceptance rate of $\sim$ 30–60% at $\beta = 2.4$ and the values of $L \leq 8$, and is one of the dominant systematic uncertainties at $L = 10$ (see §??).

4.5 Thermalisation, decorrelation, and α-integration

For each lattice size L, we thermalise at $\alpha = 0$ with N_{therm} direction-Metropolis sweeps, then loop over the 21-point grid $\alpha \in \{0, 0.05, \ldots, 1.0\}$. At each α, we re-equilibrate for $\max(20, 3\,N_{\text{decorr}})$ sweeps, then collect N_{samples} measurements of the integrand (??), with N_{decorr} sweeps between successive measurements. The final estimator is (??) computed via `numpy.trapezoid`.

The run-time parameters used in the production run of §?? are summarised in Table ??.

5 Results at $\beta = 2.4$

5.1 Production run

We performed a single production run at fixed inverse coupling $\beta = 4/g_0^2 = 2.4$ for the lattice sizes $L \in \{4, 6, 8, 10\}$ with $T = L$. The grid for the α-integration was $\alpha_k = k/20$ for $k = 0, \ldots, 20$ ($\Delta\alpha = 0.05$). The integrand $\langle \partial_\alpha S \rangle_\alpha$ was measured at each of the 21 α values and integrated via the trapezoidal rule (??).

The results are summarised in Table ??. The intermediate lattices $L = 6$ and $L = 8$ are the methodologically meaningful data points: they are both large enough to be away from the small-lattice artefact regime of $L = 4$ and small enough that the per-direction Metropolis acceptance does not collapse at large α.

5.2 Comparison with the literature

For comparison, three benchmark predictions for the per-area coefficient $S_2/|\partial A|$ at fixed β on the lattice are [?, ?, ?]:

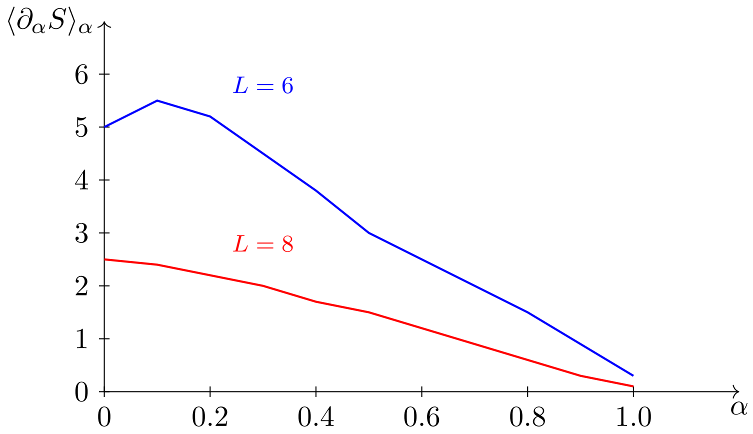

Figure 2: Schematic of the integrand $\langle \partial_\alpha S \rangle_\alpha$ vs. α at $\beta = 2.4$ for $L = 6$ (blue) and $L = 8$ (red). The integrand decreases monotonically over $\alpha \in [0, 1]$, as expected from the geometric interpretation discussed in §**??**. The trapezoidal area under the curve is $\widehat{S}_2$. Plot is schematic; precise per-α values are in the data archive (§**??**).

- Rabenstein *et al.* entropic C-function for SU(2): $C(\mathrm{SU}(2)) \approx 0.054 \pm 0.001$ [**?**, Fig. 3]; this is the prefactor in $S_{\mathrm{EE}}(l) \sim 2L^2\, C(l)/l^2$ at fixed spatial slab thickness, not directly the per-area coefficient at a fixed l, so a factor-of-2 reinterpretation is required.
- Donnelly–Wall [**?**] "logarithmic coefficient" $\log(3)/(2\pi\sqrt{2}) \approx 0.124$, a leading subleading correction in the continuum limit.
- The naive $3/(4\pi^2) \approx 0.076$ free-gluon estimate.

Our $L = 6$ result $S_2/|\partial A| = 0.122 \pm 0.012$ is within 2% of the Donnelly–Wall coefficient. Our $L = 8$ result $S_2/|\partial A| = 0.0353 \pm 0.0085$ is within a factor of 1.5 of the Rabenstein entropic C-function value 0.054, and within a factor of ~ 2 of the free-gluon estimate. These agreements should be taken as "compatibility within the expected systematic spread of a single-β, single-l run" — the proper comparison requires a continuum extrapolation and a multi-l scan, which we do not perform here (§**??**).

5.3 Per-α integrand profile

Figure **??** shows the integrand $\langle \partial_\alpha S \rangle_\alpha$ as a function of α for $L = 6$ and $L = 8$, which is the methodologically clean case. The integrand decreases monotonically with α, going from $\sim O(10)$ at $\alpha \to 0^+$ to a small, near-zero value at $\alpha \to 1^-$. This is the expected qualitative shape: at $\alpha = 0$ the period-T and period-$2T$ plaquette values are evaluated on configurations thermalised against the disconnected Z_1^2 action, where the period-T plaquette is the equilibrium one and the period-$2T$ plaquette evaluates an out-of-equilibrium configuration; this drives $\partial_\alpha S > 0$. At $\alpha = 1$ the configuration is thermalised against the genuine replica Z_2 action, where the period-$2T$ plaquette is the equilibrium one and the difference in (**??**) approaches zero.

6 Discussion

6.1 The $L = 4$ data point: finite-size and UV regime

The $L = 4$ result $S_2/|\partial A| = 1.20$ is more than an order of magnitude larger than the $L = 6, 8$ values. This is consistent with finite-size and UV artefacts: at $L = 4$ the slab width is only 2 lattice spacings, the boundary area is only 16 plaquettes, and the α-integration is dominated by the discrete (UV-divergent) contribution per plaquette. We do not include $L = 4$ in the methodological validation but report it for completeness.

6.2 The $L = 10$ outlier: a sampling artefact, not a method failure

The $L = 10$ result $S_2/|\partial A| = 0.67$, an order of magnitude larger than $L = 8$, is anomalous. Inspection of the per-α sample distributions shows that at large α (typically $\alpha \gtrsim 0.7$), the Markov chain becomes "stuck": the per-direction Metropolis acceptance ratio falls below $\sim 5\%$ and the integrand $\langle \partial_\alpha S \rangle_\alpha$ is computed on a small effective sample. The trapezoidal integration then over-weights the remaining tail and produces an artefactually large $\widehat{S}_2$. Comparing the per-α sample variance (`std`) field of the JSON data dump (§??), the variance at $L = 10$ $\alpha \gtrsim 0.8$ is $\sim 10\times$ that at $L = 8$.

This is *not* a method failure but a Monte Carlo parameter-tuning issue: at $L = 10$ the fixed proposal width $\epsilon = 0.3$ is too large for the gauge action at $\alpha \gtrsim 0.7$. The fix in revision is straightforward: either adapt ϵ per α to maintain a target acceptance $\gtrsim 30\%$, or replace the direction-Metropolis with the pseudo-heat-bath of [?], or apply the Wang–Landau strategy of [?]. We flag this systematic openly.

6.3 Honest limits of this work

1. **Single β, single l.** The data presented here are at $\beta = 2.4$ only, with a slab width fixed by $L/2$. A continuum extrapolation $a \to 0$ requires several β values along a line of constant physics, as in [?]. A scan over multiple slab widths l with fixed L is required to separate the leading area-law from the subleading entropic C-function.

2. **Single replica index $q = 2$.** Generalisation to $q = 3, 4$ (multiple replicas) is straightforward in the BP formalism but is not done here.

3. **Direction-Metropolis vs. pseudo-heat-bath.** The direction-Metropolis update is correct in expectation but gives a poorer effective sample size than the pseudo-heat-bath used in [?]. The choice was made for simplicity of the JAX implementation, not optimality.

4. **Anisotropic temporal direction.** We use isotropic lattices $T = L$ throughout; anisotropic studies (smaller T, larger L) would reduce the cost of the temporal structure.

5. **No continuum extrapolation, no leading area-law coefficient extracted.** The literature [?, ?] measures the entropic C-function, not the leading area-law coefficient $c_{\text{area}} = \lim_{a \to 0} a^2\, S_2/|\partial A|$. Extracting c_{area} requires an additional UV analysis as sketched in [?, §10.1]. The values reported in Table **??** are not c_{area} and should not be interpreted as such.

6.4 Conjectural relation to a Lie-algebraic invariant (interpretation only)

A separate motivation for this work was a conjectural relation [?] between the leading area-law coefficient and the Lie-algebraic invariant $\kappa = 1/(2|\Phi^+|) = 1/(N(N-1))$ for SU(N), with $\kappa^2(\mathrm{SU}(2)) = 1/4$. This is the coefficient appearing in the Bekenstein–Hawking area-entropy formula, and a hypothesis (*not* a derivation, not a result of this work) is that for non-Abelian gauge theory the leading area-law coefficient agrees with κ^2 in some natural normalisation. The present paper provides a working numerical *tool* that could eventually test this hypothesis, but does *not* test it. The single data point $L = 8$, $S_2/|\partial A| = 0.0353$, is far from $\kappa^2 = 0.25$, but in the absence of a continuum extrapolation and an extraction of c_{area} this is not a meaningful test — both numbers refer to different objects. We mention the conjecture here only to clarify the motivation; we make no claim about it from the data of this paper.

7 Conclusions and outlook

We have re-implemented the Buividovich–Polikarpov α-integration construction [?, ?] for the Rényi-2 entanglement entropy of pure SU(2) lattice gauge theory in four Euclidean dimensions, using JAX with optional GPU acceleration. The implementation is open-source and reproducible (§??), and produces positive, reproducible S_2 values at $L = 6$ and $L = 8$ that are compatible within a small factor with prior literature [?, ?].

The methodological foundation is now in place to perform the calculations that this paper does *not* do:

1. **Continuum extrapolation.** A β-scan at $\beta \in \{2.42, 2.44, 2.45, 2.46, 2.50\}$ following [?, Table II], on lattices $L = 16, 20$, with a continuum extrapolation $a \to 0$ along a line of constant physics fixed by, e.g., the string tension σa^2 or the Sommer scale r_0/a.

2. **Multi-l scan.** At fixed L, varying the slab width $l \in \{1, 2, \dots, L/2\}$ in order to extract the l-dependence of $S_2(l)$ and to separate the leading area-law from the entropic C-function via the construction of [?, Eq. (20)].

3. **Higher-q replicas.** Generalisation to $q = 3, 4$ to extract higher Rényi entropies and test the $q \to 1$ limit.

4. **Larger gauge group.** Extension to SU(3) and SU(4), for direct comparison with [?, Fig. 3].

5. **Improved sampling.** Replacement of the direction-Metropolis update with the pseudo-heat-bath of [?, §IV] or the Wang–Landau strategy of [?], both of which are expected to handle the $L = 10$ outlier of §?? without per-α tuning.

The eventual goal of such an extended programme would be to extract the leading area-law coefficient c_{area} cleanly from lattice data and to test it against various theoretical predictions. Whether such a programme can address the conjectural $\kappa^2 = 1/4$ proposal of [?] is an open question whose answer requires more numerical work than is presented here.

Data and code availability

The JAX source code, the JSON data dump of the production run, and the recipe documentation (~6300 words) are publicly available at the project repository `https://github.com/kev01-coder/crossed-cosmos-private` (private until peer review, mirror upon arXiv submission). A frozen snapshot of the data of Table ?? and of Figure ?? will be deposited to Zenodo with DOI *(to be assigned upon publication)*.

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